

AI-DRIVEN TEA QUALITY AND PRODUCTION IMPROVEMENTS IN MID-COUNTRY SRI LANKA

Project ID: 25-26J-133

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specializing in
Data Science

Department of Information Technology

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Accurate and timely prediction of tea yields is vital for improving productivity, profitability, and informed decision-making among tea estate managers and smallholders in Sri Lanka's mid-country agro-ecological zones. This study proposes a data-driven decision-support framework that combines historical yield information, estate-specific characteristics, environmental factors, and management practices to generate reliable and practical yield insights. The framework is built around three main components: a Monthly Yield Prediction Module that forecasts production trends, a Potential Yield Estimation Module that helps set achievable estate-level targets, and a Yield Log Book Data Extraction Module that converts traditional handwritten records into structured digital formats such as Excel. Together, these components enable accurate forecasting, efficient data preservation, and improved planning of estate resources. A user-friendly web-based dashboard allows estate managers and smallholders to enter data, visualize forecasts, assess potential yields, and access digitized logbook information. By integrating these AI-powered tools, the system promotes data-driven management, supports sustainable cultivation practices, and enhances the long-term economic resilience of Sri Lanka's tea industry.

Keywords – Tea yield prediction, potential yield estimation, yield log book data extraction, machine learning, decision-support system, tea estate management

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List of Abbreviations

Abbreviation	Description
AEZ	Agro-Ecological Zone - A region defined by specific climatic, soil, and topographical conditions that influence crop cultivation.
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
LSTM	Long Short-Term Memory
CNN	Convolutional Neural Network
VP	Vegetative Propagation
N	Nitrogen
P	Phosphorus
K	Potassium
SNN	Spiking Neural Networks
XGBoost	Extreme Gradient Boosting
RNN	Recurrent Neural Network

1. INTRODUCTION

The tea industry is a cornerstone of Sri Lanka's economy and cultural identity, with its products renowned worldwide under the "Ceylon" brand. This sector sustains millions of livelihoods, especially those of smallholders—independent, small-scale farmers who contribute over 75% of the country's tea production—and estate managers. Together, they form the backbone of Sri Lanka's tea economy, operating across diverse agro-ecological zones, including the strategically important Mid Country region. The Mid Country, positioned at elevations between 600 and 1,200 meters, offers moderate rainfall, acidic well-drained soils, and cool temperatures that uniquely influence tea growth and quality. These natural conditions demand localized cultivation strategies and yield forecasting to maximize productivity and support sustainable practices. Given the economic importance of tea exports as a source of foreign exchange and rural employment, especially among smallholders who depend heavily on the crop for their livelihoods, improving support for these growers is essential for national development and social progress.

Despite its critical role, Sri Lanka's tea cultivation faces multiple challenges including climate change impacts like erratic rainfall and temperature fluctuations, soil nutrient depletion, pest and disease outbreaks, labor shortages, and limited access to technical resources for many small-scale farmers. Traditional yield prediction methods, relying on historical averages, often fall short in providing accurate, real-time guidance for farmers, particularly in capturing the dynamic environmental and agronomic conditions of specific regions like the Mid Country. Another major challenge is that most tea estates maintain yield records in manually written yield log books, which are vulnerable to damage and loss. Scanning and digitizing these records can preserve valuable historical data and enable more accurate, data-driven yield predictions. By delivering actionable, data-driven insights tailored to smallholders and estate managers, the study seeks to enhance decision-making, help set realistic yield targets, and improve economic outcomes. Ultimately, this approach aspires to strengthen the sustainability and resilience of Sri Lanka's tea industry, empowering local growers to adapt to changing conditions and contribute effectively to the country's agricultural and socio-economic future

1.1 Background & Literature Survey

The global tea industry sustains a multibillion-dollar trade, with annual production surpassing billions of kilograms to meet the demands of diverse markets worldwide [14]. Beyond its economic scale, tea fosters international connections through the export of specialty varieties and drives development in producing economies by generating critical foreign exchange [14]. In Sri Lanka, often referred to as “Ceylon” in the global tea market, tea cultivation and export have profoundly shaped the nation’s identity, providing employment to over a million people directly and indirectly [12], [13]. While large tea estates dominate the country’s central regions, numerous smallholders scattered across rural areas depend heavily on tea cultivation as their primary source of livelihood [10], [13]. Revenue from tea exports remains vital for Sri Lanka’s economic resilience and national development, funding infrastructure projects, public services, and contributing to social progress [10]. As such, the tea industry continues to be central not only to Sri Lanka’s agricultural economy but also to the fabric of its rural communities and its stature in the global market [13].

Tea cultivation in Sri Lanka faces critical challenges that threaten both productivity and sustainability. Climate change has caused erratic rainfall and unpredictable temperatures, directly affecting crop growth and yield stability [14]–[17]. Soil nutrient depletion from continuous tea monoculture without adequate replenishment weakens plant health, while persistent pest and disease outbreaks—such as blights and root rots—reduce both yield and tea quality [12]. Smallholders, who contribute the majority of Sri Lanka's tea production, often have limited access to modern agricultural knowledge, resources, and extension services, hindering their ability to adopt improved practices [13]. Additionally, the costly and slow process of replanting aging tea bushes hampers efforts to sustain high yields [12]. Labor shortages and rising wages further add economic pressures on both smallholders and estate managers [10]. These interconnected challenges underline the urgent need for targeted support through data-driven yield prediction and decision-making tools, which this project aims to develop using unique insights from the Mid Country region to empower growers and enhance productivity [8], [9], [11].

In Sri Lanka, most tea estates still record yield information in handwritten logbooks. Over time, these physical records are prone to damage, fading, or even complete loss, which can erase valuable historical data. The absence of digitized versions makes it difficult to access and analyze information efficiently, creating a major obstacle to introducing modern, data-driven decision-support systems within the tea industry. Therefore, preserving and converting these traditional records into digital formats is crucial for maintaining data accuracy, enabling detailed yield analysis, and supporting long-term planning.

Tea cultivation in Sri Lanka is shaped by a diverse mosaic of agro-ecological regions, each defined by its unique climate patterns, soil composition, and topographical features [12], [13]. These regions—ranging from low country plains to highland plateaus—directly influence the suitability for tea growth, the productivity of plantations, and the characteristics of the resulting tea [12], [14]. Figure 1.1 illustrates all the agro-ecological regions of Sri Lanka, while this research specifically focuses on the Mid Country region. The Mid Country agro-ecological region, situated at elevations between 600 and 1,200 meters above sea level, is renowned for its moderate rainfall, well-drained acidic soils, and relatively cool temperatures [12], [13]. These natural conditions create a distinct environment where tea plants flourish, leading to unique yield potentials and flavor profiles that distinguish Mid Country teas from those of other zones [13]. Importantly, the performance of tea plants can vary substantially across Sri Lanka's different agro-ecological zones, making it essential to tailor research and management strategies to specific local conditions [12], [16], [17]. By focusing exclusively on the Mid Country region, this research ensures that predictive modeling and recommendations are grounded in the environmental realities faced by growers in this critical segment of Sri Lanka's tea industry [8], [9], [11].

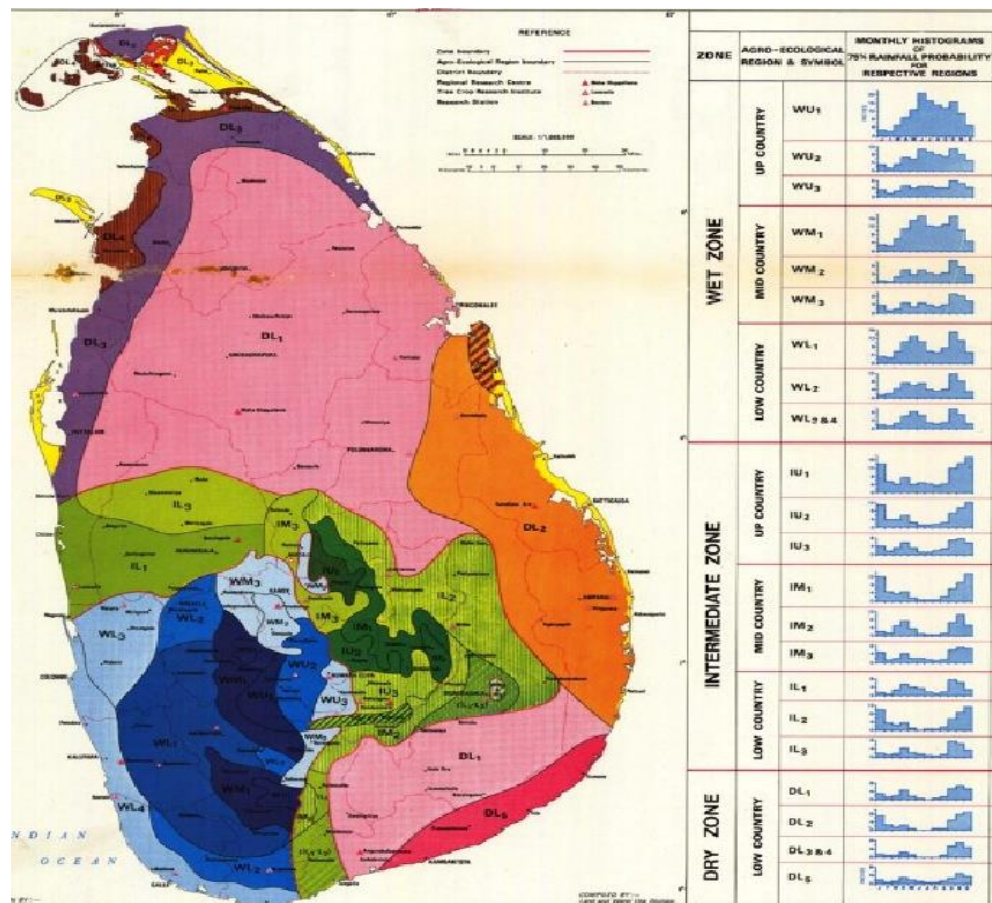


Figure 1.1: Agro Ecological Regions of Sri Lanka

Environmental factors play a crucial role in determining tea plant growth and productivity, affecting both the quantity and quality of yields [12]. Numerous studies have identified rainfall as a key variable, with optimal tea production requiring adequate and well-distributed precipitation; however, both excessive and insufficient rainfall can negatively impact yields by causing moisture stress or nutrient leaching [4], [16]. Essential soil nutrients—particularly nitrogen (N), phosphorus (P), and potassium (K)—directly influence shoot growth and leaf quality [12]. The age of the tea bush also significantly affects yield: younger bushes produce lower yields that increase with maturity, while older bushes may experience productivity declines without proper management such as timely pruning [12]. Pruning influences the re-growth cycle and the vigor of new shoots, which are critical for leaf harvesting [12]. Moreover, factors like vegetative propagation (VP) or seedling type impact plant vigor

and adaptability to local environmental conditions [6], [7]. These environmental and physiological variables interact in complex ways that require careful monitoring and integration to enable accurate crop yield predictions and improved cultivation practices [8], [9], [11]. Recent research emphasizes the value of incorporating such agro-environmental factors into machine learning models, which can enhance the accuracy of yield forecasting under diverse climatic and soil conditions [2], [3], [17].

Currently, estate managers often set monthly yield targets by averaging data from the previous five years. While this method offers a straightforward benchmark, it does not account for important environmental factors like rainfall, and soil nutrients, which vary over time and significantly impact tea yields [12], [16]. This research aims to improve yield prediction accuracy by incorporating these historical environmental factors along with pruning year data [8], [9]. By using comprehensive historical data on climate and soil conditions, the study seeks to provide estate managers and smallholders with more precise yield targets and better decision-making tools to optimize tea production [8], [9], [11].

Several studies have explored tea yield prediction using a range of methodologies, highlighting the growing role of machine learning and AI in optimizing production. In Sri Lanka, explainable AI techniques have been applied to estimate Ceylon tea crop yields, incorporating environmental variables and historical data to improve prediction accuracy [8]. In Pakistan, multimodal data fusion combined with deep neural networks has been employed, integrating satellite imagery, NDVI, and weather data to achieve highly accurate yield estimates [9]. Hybrid approaches that combine simulation models with machine learning, such as AquaCrop and XGBoost, have also demonstrated effectiveness in predicting tea yields [11]. Earlier work has investigated the influence of rainfall and temperature on tea yield in North-East India, emphasizing the critical role of climatic factors [4]. Random forest models have been successfully used to forecast tea production in Bangladesh over multi-decadal periods, achieving high predictive performance [5]. Other machine learning approaches, including spiking neural networks (SNN) and regression-based techniques, have been applied to crop yield prediction in the broader South Asian

context [2], [3]. Collectively, these studies underscore the importance of integrating climatic, soil, and agronomic variables into predictive models, forming a strong foundation for developing more precise and practical yield prediction systems for tea plantations.

1.2 Research Gap

Despite recent advancements in tea productivity modeling, significant gaps remain in developing predictive frameworks tailored to specific agro-ecological zones. Current models primarily employ generalized approaches without accounting for the unique environmental and management factors of distinct regions, particularly the Mid Country agro-ecological zone. Furthermore, existing studies have not integrated all pivotal factors influencing tea yield, including pruning year, soil nutrient status, and propagation methods, into a comprehensive model. Additionally, there is a notable absence of predictive systems that estimate potential yield to assist estate managers and smallholders in setting realistic targets. Moreover, most tea estates in Sri Lanka still maintain yield records in manually written yield log books, which are prone to damage and loss; scanning and digitizing these records would preserve valuable historical data and enable more accurate, data-driven yield predictions. These gaps underscore the need for a localized, multi-factor, and decision-oriented predictive model tailored specifically to the Mid Country agro-ecological zone.

Key Research Gaps:

1. Agro-Ecological Zone (AEZ) Based Yield Prediction

Existing research on tea yield prediction does not account for AEZ-specific conditions. There is a lack of models that provide monthly and annual forecasts tailored to local environmental realities.

2. Potential Yield Target Estimation

Conventional approaches for setting estate yield targets rely on simple historical averages and do not estimate the potential yield an estate can achieve. There is a gap in predictive systems that can provide data-driven potential yield targets.

3. Data Digitization and Accessibility

Most tea estates in Sri Lanka still maintain yield records in manually written yield log books, which are highly susceptible to physical damage, loss, and deterioration. There is a significant gap in the digitization of these records, and no established

framework exists for systematically scanning and converting yield log book data into digital formats suitable for analysis. This lack of accessible, digitized historical yield data limits the development and validation of advanced predictive models and decision-support systems.

4.Integrated Decision-Support System

Most existing systems handle yield prediction separately from management recommendations. There is a need for a unified framework combining AEZ-based yield forecasts, potential target estimation, and logbook data extraction to support estate decision-making.

Conclusion:

There is a clear need for an integrated, data-driven decision-support system for the tea industry that goes beyond traditional yield prediction methods. Current approaches do not adequately capture AEZ-specific conditions, provide realistic potential yield targets, or leverage manually maintained yield log books, which if scanned and digitized, can preserve valuable historical data for more accurate predictions. By addressing these gaps through a unified framework, estates can move from reactive to proactive decision-making, enabling more accurate forecasting and ultimately improved sustainability.

Table 1.1: Different Between Existing System and Proposed System

Feature	Existing Research	Proposed System	Research Gap
System Integration	Most studies focus on either environmental modeling ([12], [14], [16]) or machine learning-based prediction ([2], [3], [5], [11]); no integration of temporal + static agronomic factors	A unified predictive framework that integrates climatic, agronomic, and management variables for Mid Country tea.	Lack of integrated systems that consider all influencing factors together
Technological Approach	Approaches include regression [12], CNN/VGG-16 [7], Spiking Neural Networks [3], and simulation-ML hybrids [11]. Each method has limitations; mostly tested in controlled settings.	AI/ML-based predictive model tailored to Mid Country tea, designed for field applicability.	Current methods are fragmented; no specialized predictive system designed for localized agro-ecological tea yield forecasting.
Data Source & Sampling	Data often taken from one factory (e.g., Dhammapla [5]) or a few elevations [12], [13]. Some use 2–3 regions but ignore agro-ecological diversity.	Stratified sampling from all agro-ecological regions in the Mid Country zone, combining	Prior studies lack representative sampling; no dataset fully reflects Mid Country variability.

		estate-level datasets into a representative dataset.	
Tea yield Log Book Image Scanning & Data Extraction	Not addressed	Capture images of manually maintained tea yield log books, process them using image processing techniques, and extract data directly into Excel	No existing framework for systematically scanning estate yield log books and converting them into usable digital data, limiting predictive model development
Real-World Application	Systems often designed for experimental purposes (e.g., smart monitoring [1], waste analysis [10]) or narrow geographic focus. Limited usability for estate managers / smallholders.	Practical, decision-support predictive system for estate managers and smallholders in Mid Country.	No operational decision-support system exists to forecast yields and assist planning at estate/smallholder level.

1.3 Research Problem

Despite the critical economic and social importance of tea cultivation in Sri Lanka, estate managers and smallholders continue to face significant challenges in accurately forecasting yields. Traditional approaches, such as averaging yields from the previous five years, provide only rough estimates and fail to account for the dynamic interactions of environmental, soil, and management factors that influence tea productivity. Variations in rainfall, soil nutrients, pruning cycles, and propagation methods contribute to substantial fluctuations in yield, yet these factors are rarely integrated into a comprehensive predictive framework. Furthermore, existing research often focuses on limited geographic areas or aggregated data across broad climatic zones, neglecting the unique agro-ecological characteristics of the Mid Country region. Another major challenge is that most tea estates still maintain yield records in manually written yield log books, which are prone to damage, loss, and deterioration. Without digitization or systematic scanning of these records, valuable historical data remains inaccessible, limiting the development and validation of advanced predictive models. This lack of localized, data-driven predictive systems limits growers' ability to make informed decisions regarding crop management, resource allocation, and economic planning, highlighting an urgent need for an accurate, region-specific tea yield prediction model that can enhance productivity and sustainability.

Yield prediction models that are tailored to specific agro-ecological regions offer estate managers and smallholders more practical and actionable tools beyond generic production estimates. By aligning yield forecasts with the unique environmental and soil conditions of each region, growers can better anticipate variations in crop performance and make more informed management decisions. Accurate regional predictions support effective labor planning by helping estates schedule plucking rounds and allocate workforce resources efficiently during peak harvest periods. In addition, they facilitate optimized nutrient management by linking expected yields to soil nutrient needs, ensuring fertilizer applications are both timely and cost-effective.

2. OBJECTIVES

2.1 Main Objective

The main objective of this research is to develop an AI-powered web application that facilitates data-driven decision-making in Sri Lanka's tea industry, with a particular emphasis on mid-country tea estates. The system is designed to provide accurate monthly tea yield predictions based on agro-ecological zones, enabling estate managers to plan operations more effectively. In addition, the platform yield log book scanning and data extraction feature, which digitizes manually written yield records into analyzable formats. Ultimately, this research aims to create a comprehensive decision-support tool that enhances operational efficiency, improves resource allocation, reduces uncertainty, and strengthens both the competitiveness and profitability of the mid-country tea sector.

2.2 Sub Objective

The following are the specific objectives that must be completed to achieve the main goal of developing the AI-powered Tea Yield Prediction Decision-Support System. This section will outline the specific goals for each component of the system in greater detail, including monthly yield forecasting, potential yield estimation, yield log book scanning and data extraction, and decision-support features for estate management.

1. Integrate Data from Mid-Country Agro-Ecological Zones

Collect and consolidate data from all agro-ecological zones within the Mid Country, ensuring that the system reflects the variability in climate, soil, and management practices. This increases the accuracy and generalizability of predictions across the region.

2. Develop a Monthly Tea Yield Prediction Model

Design and implement a predictive model that accurately forecasts monthly tea yields based on agro-ecological zones, environmental factors (rainfall, humidity, soil nutrients), and management practices (bush age, pruning cycles, propagation method). This enables estate managers to plan operations more efficiently and respond to temporal variations in yield.

3. Create a Potential Yield Estimation Module

Develop a system to estimate potential yields for estates, providing realistic production targets that go beyond traditional historical averages. This helps managers set achievable goals and identify gaps between actual and potential output.

4. Implement Yield Log Book Scanning and Data Extraction Feature

Develop a module to scan manually written tea yield log books, process images, and extract data into digital formats such as Excel. This feature preserves valuable historical records, ensures accessibility, and supports advanced data analysis for improved predictive modeling and decision-making.

5. Develop a Unified AI-Powered Web Application

Design and implement a single, integrated web platform that combines all system components—including monthly yield prediction, potential yield estimation, and logbook scanning with data extraction—into a cohesive interface. The application should allow estate managers and smallholders to input data, view real-time predictions, and access actionable recommendations, simplifying the interpretation of complex analytics for practical, data-driven decision-making.

3. METHODOLOGY

3.1 Project Overview

This project aims to develop an AI-powered decision-support system for Sri Lanka's tea industry, specifically focused on Mid-Country estates. The primary objective is to produce agro-ecological-zone (AEZ) based tea yield predictions at both monthly and annual resolutions, enabling estate managers and smallholders to plan operations according to local environmental conditions. In addition to forecasting, the system will provide potential yield targets for estates and include a yield log book scanning and data extraction module, allowing manually written tea yield records to be digitized and converted into structured formats (e.g., Excel) for improved data accessibility and predictive modeling.

To achieve these goals, the system will employ a hybrid modeling framework that integrates temporal features—such as rainfall, nutrients, plucking rounds, and past yields—with static estate features like bush age, pruning year. The approach is designed to be flexible, allowing the selection and tuning of machine learning and deep learning models (e.g., LSTM/GRU for time-series prediction and tree-based models for static features) according to data characteristics.

The resulting system will deliver monthly and yearly yield forecasts, potential production targets, and digitized historical data from scanned log books, enabling estate managers to make informed decisions regarding labor allocation, pruning schedules, and based on AEZ-specific insights. Its modular design ensures that components can be adapted or replaced as new data or techniques become available, ensuring long-term applicability and scalability across multiple estates.

3.2 System Overview Diagram

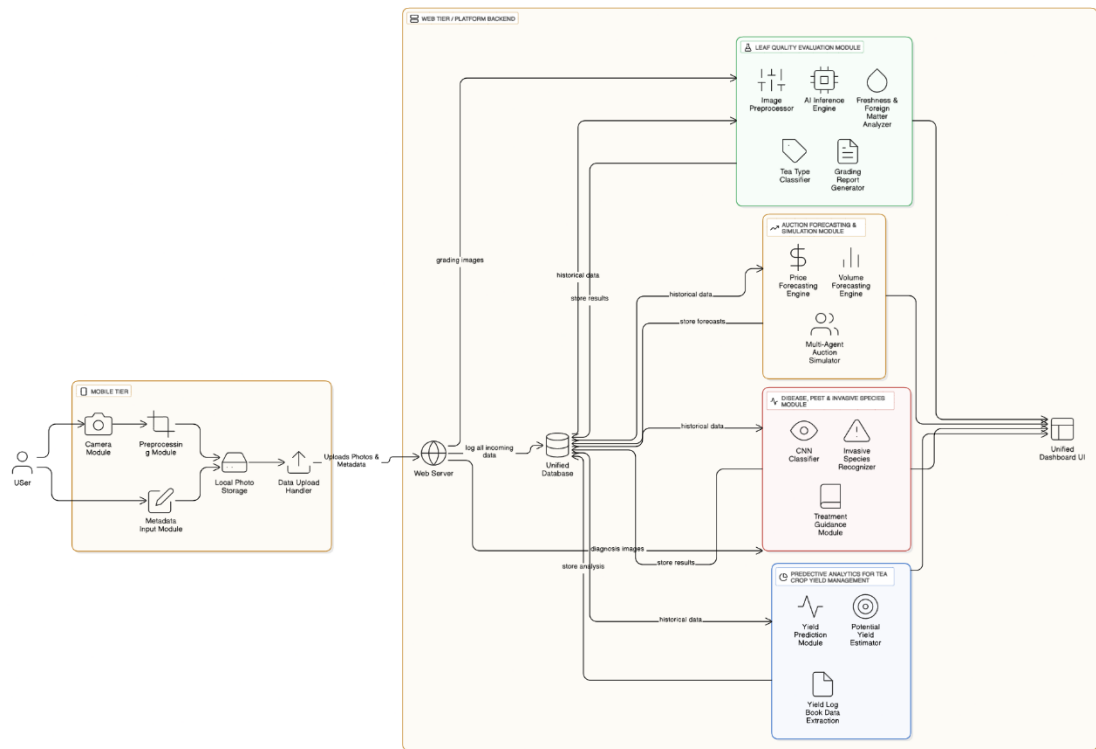


Figure 3.1 System Overview Diagram

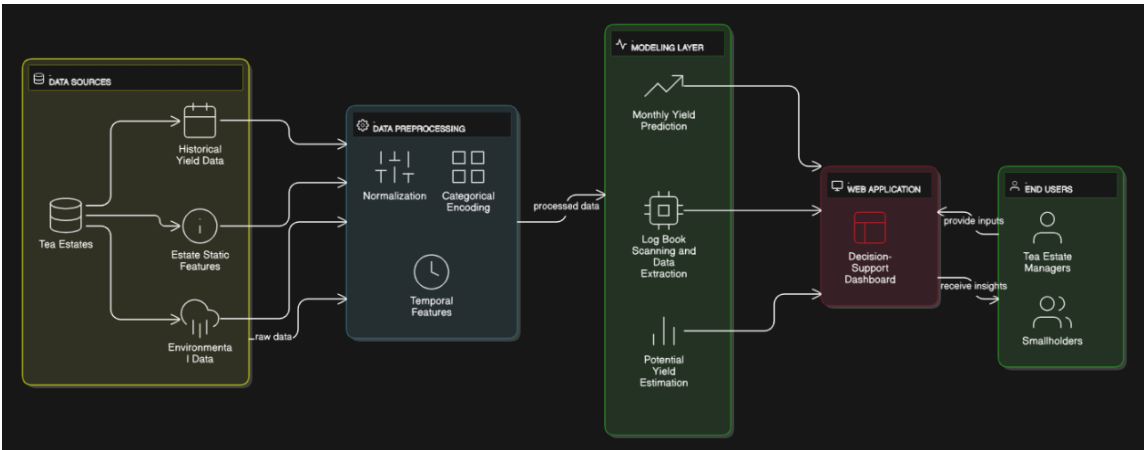


Figure 3.2: Individual Diagram

3.3 System Overview

The proposed system integrates multiple artificial intelligence–driven modules to modernize and streamline the tea supply chain, from leaf plucking to market forecasting. At its highest level (second diagram), the architecture brings together four main components:

Predictive Analytics for Tea Crop Yield Management & Log Book Scanning – historical, environmental and estate yield log book data feed ML models to forecast production outcomes and support data-driven decision-making

Automated Tea Leaf Grading & Inspection – machine vision models classify leaf quality and freshness.

AI-Based Detection of Tea Plant Diseases – deep learning systems identify signs of pest attacks or disease from leaf imagery.

Tea Market Demand Prediction & Auction Planning – time-series forecasting of auction/market data informs smarter supply planning.

All outputs feed into a unified dashboard that provides actionable insights grading data, disease alerts, yield forecasts, and demand predictions for stakeholders including farmers, factories, and industry experts.

As part of this framework, the proposed system also incorporates a data-driven tea yield prediction and yield log book scanning/data extraction module to support decision-making for tea estate managers and smallholders. It integrates historical yield records, estate-specific characteristics (pruning year), environmental factors (rainfall, soil nutrients), and management practices (pruning cycles, propagation methods, plucking rounds). After preprocessing and feature engineering, the system applies machine learning models across three key modules: a Monthly Yield Prediction Module that forecasts production trends, a Potential Yield Estimation Module for realistic estate-level targets, and a Yield Log Book Data Extraction Module that digitizes historical records into Excel or other formats. A user-friendly web application

dashboard allows managers and smallholders to input data, visualize forecasts, analyze potential targets, and interact with digitized yield log book data, empowering evidence-based decisions that maximize productivity.

3.4 Software Development Process

The software development process for the proposed AI-powered Tea Yield Prediction and Yield Log Book Scanning System is structured into iterative and manageable stages, following the Agile methodology. Agile is considered the most suitable approach for this project due to its flexibility, adaptability to evolving data requirements, and emphasis on continuous improvement—key aspects when developing machine learning models and integrating real-world estate-level data sources.

Agile promotes incremental development, enabling continuous validation and improvement at each stage. This ensures that components such as yield prediction, potential yield estimation, and yield log book scanning with data extraction are developed, tested, and refined iteratively. The process will progress through six main Agile stages: requirements gathering, analysis, design, implementation, testing, and maintenance. Each cycle will produce a working prototype that incorporates user feedback and performance improvements in subsequent iterations until the final version is fully optimized.

To effectively manage the Agile process, the project will prioritize collaboration, adaptability, and regular progress evaluation. Short development sprints, progress reviews, and feedback sessions will facilitate timely issue resolution and ensure consistent alignment with stakeholder needs. This iterative approach will help refine both the predictive models and the data extraction modules to achieve greater reliability and accuracy.

The use of Agile in this project supports efficient coordination between developers, data analysts, and domain experts from the tea industry. Given the system's evolving data inputs and analytical complexity, Agile provides the ideal framework to maintain

flexibility, ensure scalability, and deliver a **robust, accurate, and user-friendly decision-support platform** for estate managers and smallholders.

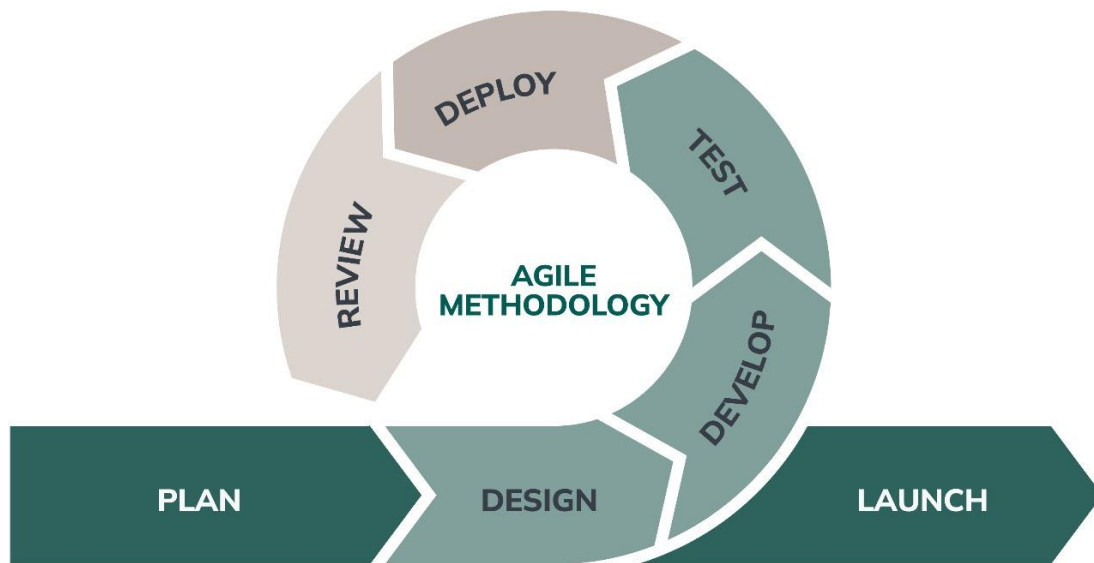


Figure 3.2 Software Development Life Cycle

3.4.1 Requirements Gathering and Analysis

The requirements gathering process for this project will follow an iterative, sprint-based approach to ensure that the system meets the dynamic and varied needs of tea estate managers and smallholders. Given the complex interactions between environmental factors, agronomic practices, and economic considerations, it is essential to continuously refine the requirements as new insights emerge from data analysis and stakeholder feedback.

In the initial phases, the focus will be on broad system functionality, such as the ability to predict monthly tea yields based on general environmental and estate-level factors. As development progresses, requirements will evolve to include more specific features, such as potential yield estimation, AEZ-specific predictions, and scanning manually recorded tea yield log books with automated data extraction into digital formats to preserve historical records and support advanced predictive modeling.

Stakeholder input will be central to this process:

- **Estate Managers and Smallholders:** To understand operational priorities, decision-making challenges, and the types of predictions and simulations most valuable for planning labor, pruning schedules, and resource allocation.
- **Agronomists and Tea Experts:** To ensure that the predictive models accurately reflect the influence of key agronomic factors, including pruning cycles, soil nutrient status, and propagation methods.

Through continuous collaboration and iterative refinement, the system requirements will be developed to align with both the technical feasibility and practical needs of end-users. This approach ensures that the resulting AI-powered platform not only provides accurate yield forecasts, automated yield log book data digitization, and actionable insights, but also supports informed estate management decisions.

3.4.2 Feasibility Study

- **Economic Feasibility**

The proposed AI-powered decision-support system for tea yield prediction, potential yield estimation, and yield log book scanning with automated data extraction is designed to deliver substantial benefits to Sri Lanka's tea sector, particularly in the mid-country estates. As a primarily software-driven solution, the system does not demand major hardware investments, making it cost-effective for both estate managers and smallholders. The main costs are expected to include system design and architecture, software development and integration of machine learning models, cloud hosting and data storage, and ongoing maintenance and support. Compared to the long-term gains of improved forecasting accuracy, resource optimization, and enhanced profitability, the investment is economically feasible and justifiable.

- **Technical Feasibility**

The system leverages advanced machine learning and AI models for time-series yield prediction, potential yield estimation, and yield log book scanning with automated data extraction, ensuring technical soundness and reliability. A hybrid modeling framework will be employed to capture both temporal and static features, making it well-suited to the structured, multi-year datasets collected from estates. The proposed web application will integrate the predictive models and the yield log book scanning module into a unified platform that is practical for estate managers and smallholders. Given the maturity of AI technologies and the availability of relevant data, the system is technically feasible and can be successfully implemented with the resources and expertise available.

- **Operational Feasibility**

Operational feasibility is strong, as the system directly addresses the pressing needs of estate managers and tea smallholders by providing accurate, localized, and actionable insights. The user-friendly web interface ensures that predictions and log book scanning outputs can be accessed in real time, supporting decision-making in areas

such as labor planning, nutrient management, and estate-level scenario evaluation. By aligning with the operational workflows of tea estates, the system minimizes disruption while adding significant value to day-to-day management practices. This alignment ensures high usability and adoption potential across stakeholders.

- **Schedule Feasibility**

The project has a clearly defined scope, with milestones set for data collection, model development, system integration, and testing. A phased implementation approach allows for iterative improvements, beginning with the yield prediction module and expanding to potential yield estimation and yield log book scanning with data extraction module. Based on the availability of historical datasets and the structured project design, the system can be developed within the proposed timeframe, making the project schedule feasible and realistic.

3.4.3 Dataset

The dataset for this project will consist primarily of historical and current tea estate data collected from estates across the Mid-Country agro-ecological zones. Data will be gathered from estate records, soil and weather monitoring systems, and market sources, ensuring a comprehensive representation of environmental, management, and economic factors affecting tea yield.

The dataset will support three main objectives:

1. Monthly Yield Prediction

The dataset for monthly yield prediction will include the following estate-level data:

- **Environmental Factors:** Rainfall and humidity recorded at the estate.
- **Soil Nutrients Applied:** Quantities of key fertilizers or nutrients (N) applied during the month.
- **Management Practices:** Information on pruning cycles, propagation methods, fertilizer application schedule, and number of plucking rounds.
- **Historical Yield Records:** Monthly and annual yield data for each estate, used as the target variable for model training.

This structure ensures that the model leverages the key factors that directly influence tea production and yield trends.

2. Potential Yield Estimation

Data will capture the maximum achievable yield under optimal agronomic and environmental conditions, allowing the system to estimate realistic production targets beyond historical averages. This will include information on estate-specific management capacity, soil fertility potential, and historical performance trends.

3. Yield Log Book Scanning and Data Extraction

The dataset for this module consists of scanned images of manually written tea yield log books. These images serve as the primary data source. Using image processing techniques and optical character recognition (OCR), the information from these log

book images will be systematically extracted and converted into digital formats such as Excel. This process preserves valuable historical records, ensures data accessibility, and provides structured input for advanced predictive modeling and decision-support analyses.

Dataset Characteristics

- **Diversity:** Data will be collected from multiple estates spanning different elevations, soil types, and microclimates within the Mid-Country zone. This ensures the dataset reflects AEZ variability and supports robust model generalization.
- **Annotation:** All data points will be standardized, cleaned, and validated for accuracy. Missing values will be handled using appropriate imputation techniques, and categorical variables such as AEZ will be encoded for model training.
- **Splitting:** The dataset will be divided into training (70%), validation (15%), and testing (15%) subsets to facilitate balanced model development and unbiased evaluation.
- **Integration with External Sources:** Where available, publicly accessible tea industry datasets, meteorological records, and soil surveys will be incorporated to enhance model accuracy and coverage.

3.4.4 System Implementation

The system will be developed using a combination of data engineering, machine learning, and web technologies to provide actionable yield predictions, potential yield estimation, and perform logbook scanning and structured data extraction.

1. Data Collection and Integration

- Gather historical and current data from estates, including monthly yields, soil nutrients applied, climate data, and management practices.
- Collect scanned images of manually written tea yield log books for digitalization.
- Standardize and clean data to handle missing values, inconsistencies, and outliers.

2. Feature Engineering

- Identify key features impacting yield: rainfall, soil nutrients, pruning cycles, propagation methods.
- Encode categorical variables (e.g., AEZ) and normalize continuous variables for machine learning models.

3. Model Development

- Develop a hybrid predictive framework combining:
 - Time-series models (LSTM/GRU): Capture temporal patterns in monthly yield data.
 - Tree-based models (Random Forest, XGBoost): Handle static and agronomic features.
- Train separate models for:
 - Monthly yield prediction.
 - Potential yield estimation under optimal conditions.
 - Use preprocessed data extracted from scanned log books as input to predictive models.
- Use training (70%), validation (15%), and testing (15%) datasets for model development and evaluation.

4. Evaluation Metrics

- Assess regression models with MAE, RMSE, and R^2 to measure prediction accuracy.
- Perform cross-validation and sensitivity analysis to ensure robustness across estates.

5. System Deployment

- Develop a web-based application with an intuitive interface for estate managers and smallholders.
- Functionalities include:
 - Input estate-specific data (pruning cycle, historical yields).
 - Generate monthly yield predictions.
 - Estimate potential yields for realistic target setting.
 - Upload log book images and extract data into digital formats for analysis.
- Support real-time data updates, interactive dashboards, and actionable visualizations for informed decision-making.

3.4.5 Testing

The testing phase ensures that the Tea Yield Prediction and Yield Log Book Scanning/Data Extraction System functions correctly, produces accurate predictions, and meets both functional and non-functional requirements. The goal is to verify that the system provides reliable yield forecasts, potential yield estimates, and successfully scans log book images to extract data for analysis.

Unit Testing:

- Test individual components such as data preprocessing, feature engineering modules, machine learning models for yield prediction, potential yield estimation, and log book scanning and data extraction modules.
- Verify that each module produces correct outputs given sample inputs and handles edge cases appropriately.

Integration Testing:

- Validate the interaction between system components, including data ingestion, preprocessing pipeline, predictive models, log book scanning module and the web interface.
- Ensure seamless flow of data from estate inputs to final predictions and scenario simulations.

User Acceptance Testing (UAT):

- Conduct sessions with estate managers and smallholders to assess usability, clarity of predictions, and usefulness of log book scanning and extracted data integration.
- Collect feedback to ensure the system supports practical decision-making in real-world estate operations.

System Testing:

- Validate that all features work cohesively and deliver consistent results.

Performance Testing:

- Measure the efficiency of model predictions, potential yield computations, and log book scanning/data extraction processing.
- Ensure the system handles multiple estates and large datasets without significant delays.

Security Testing:

- Verify that user data, including estate records, scanned log book images, and input parameters, are stored and transmitted securely.
- Test authentication, access control, and data encryption mechanisms to protect sensitive information.

3.4.6 Work Breakdown Structure

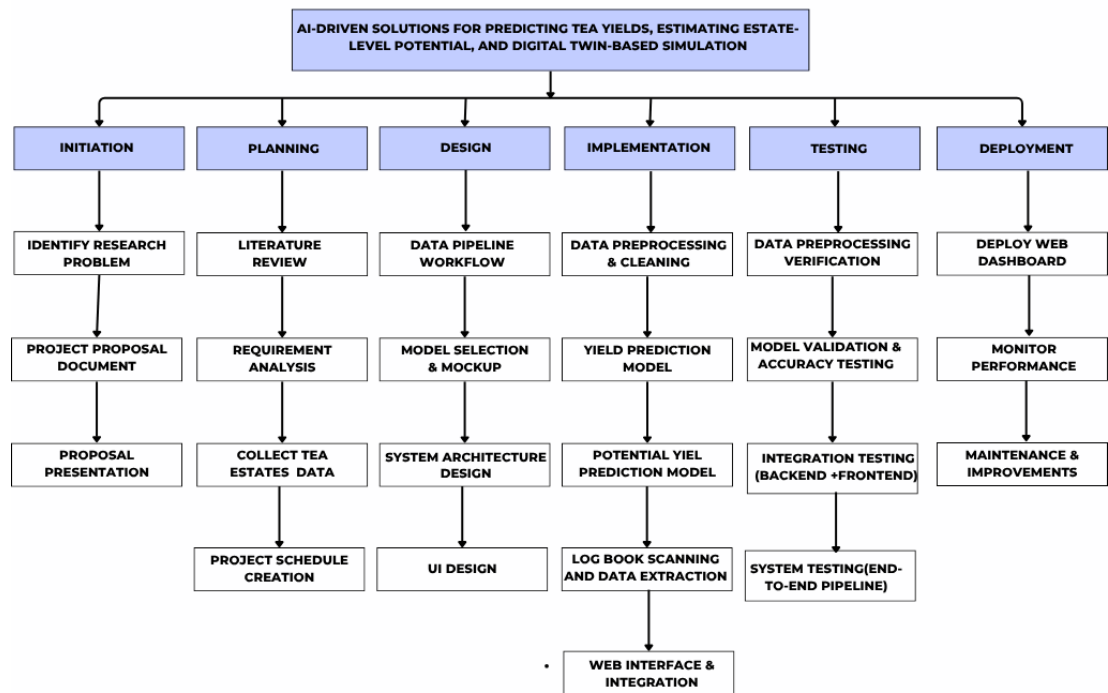


Figure 3.3 : Work Breakdown Structure

3.4.7 Gantt Chart



Figure 3.4 : Gantt Chart

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

In this research component, the functional requirements are given below.

- **Login to the System** – Estate managers and authorized users must log in to the system using a valid username and password to ensure secure access.
- **Monthly Tea Yield Prediction** – The system will predict monthly tea yields for each estate by combining temporal features (rainfall, nutrient levels, past yields, plucking rounds) with static estate features (bush age, pruning history, VP/seedling type). Results will be displayed with accuracy metrics and uncertainty ranges.
- **Potential Yield Estimation and Target Setting** – The system will generate conservative (baseline) and aspirational (potential) yield targets for each estate. These targets will be based on seasonal variations, pruning cycles, and nutrient/climate conditions to support data-driven planning.
- **Yield Log Book Scanning and Data Extraction**– The system will scan manually written tea yield log books, process the images using OCR and image processing techniques, and extract relevant data into digital formats such as Excel. This ensures historical records are preserved, accessible, and ready for integration with predictive models.
- **View Estate Reports and Insights** – Users can view detailed reports including predicted monthly yields, potential targets, yield log book data summaries, confidence intervals, and the main drivers influencing results.
- **User-Friendly Web Interface** – A web-based platform will be developed to allow managers to input estate details, view results in real-time, and download reports for decision-making.

- **Data Storage and Management** – The system will securely store estate data, climate information, predictions, scanned log book data, and reports in a centralized database for easy access, management, and future model improvements.

4.2 Non-Functional Requirements

In this application, non-functional requirements must also be considered to ensure that the solution is reliable, efficient, and user-friendly for estate managers and other stakeholders. The system is designed with an intuitive and interactive interface to allow users to easily input estate and climate data, scan log book images, extract relevant data into digital formats, view predictions, and make informed decisions without technical difficulty. In addition to usability, there are several other key non-functional attributes as described below.

- **Security**

The system will ensure that all estate data, including historical yields, climate variables, and management records and digitized log book data, are securely stored and accessible only to authorized personnel. This protects sensitive estate and operational information.

- **Availability**

The system will be accessible whenever estate managers or agronomists need to view predictions or set potential yield targets, or upload and process log book images, ensuring timely decision-making for monthly operations.

- **Usability**

The web application will be designed for ease of use, allowing estate staff to input data, scan log books, view predicted yields, and analyze trends without requiring advanced technical knowledge.

- **Accuracy**

The system will aim to provide reliable yield predictions and potential target estimates, and accurate extraction of log book data using OCR techniques, leveraging

both temporal and static features to ensure robust results that reflect estate-specific conditions.

- **Performance**

The system will be optimized to process multi-year monthly data efficiently, handle large batches of scanned log book images, and delivering predictions quickly to support monthly planning and operational decisions.

4.3 User Requirements

- **Easy account registration and login** – Users expect a simple, streamlined process for account creation and secure login.
- **Provides secure login access** – Users expect their estate data, including digitized log book data and reports to be stored securely with access controls.
- **Accurate and reliable predictions** – Users expect high accuracy in monthly yield forecasts, potential targets, and data extracted from scanned log books.
- **Clear reporting and visualization** – Users expect easy-to-understand reports, charts, and tables for decision-making.
- **User-friendly interface** – Users expect a simple and intuitive platform to enter data, scan and upload log book images, access predictions without technical complexity.
- **Fast system response** – Users expect minimal response time when generating predictions, reports, or processing log book scans.
- **Transparency and explainability** – Users expect to see the key drivers behind predictions and understand how log book data was interpreted, ensuring trust in the system.

4.4 Software Requirements

- **User-end**

Web browser

Ability to upload scanned images of tea log books for data extraction

- **Developer-end**

Programming Languages – Python v3.8 or above, JavaScript (ReactJS 18)

Frameworks – Flask 2.4 or above / Streamlit for backend integration

Deep Learning Libraries – TensorFlow v2.13 or PyTorch 2.0 for LSTM/GRU models

ML Libraries – XGBoost v1.7 or above, LightGBM v3.3 or above, Scikit-learn

OCR and Image Processing Libraries – OpenCV, Tesseract OCR (for scanning and extracting log book data)

Data Handling – Pandas, NumPy

Visualization – Matplotlib, Seaborn, Plotly

Version Control – Git and GitHub

4.5 Test Cases

Table 4.1: Test Case 1

Test case ID: MID_Test_01			Test designed by: Waduthanthri W.T.H.D		
Test title: Monthly Yield Prediction			Test designed day: 2025/8/28		
Test priority (High/Medium/Low): High			Test executed by: Waduthanthri W.T.H.D		
Module name: Yield Prediction			Test executed day: 2025/12/28		
Description: Predict monthly tea yield for selected estate					
Pre-conditions: User logged in and estate selected					
Testing steps: 1. Login to the system as manager 2. Navigate to the main menu. 3. Go to the Predict Yeild Page 4. Click “Predict Yield” button.					
Test ID	Test Inputs	Expected Output	Actual Output	Result (Pass/Fail)	Comments
MID_Test_01	Input estate and month/year along with relevant climatic, nutrient, and agronomic features for yield prediction.	Predicted monthly yield (kg).	Predicted monthly yield (kg)	Pass	Predicted monthly yield (kg) Accurate prediction is generated by the model.

Table 4.2: Test Case 2

Test case ID: MID_Test_02			Test designed by: Waduthanthri W.T.H.D		
Test title: Potential Yield Estimation			Test designed day: 2025/8/28		
Test priority (High/Medium/Low): High			Test executed by: Waduthanthri W.T.H.D		
Module name: Target Setting			Test executed day: 2025/12/28		
Description: Generate baseline & potential targets for an estate					
Pre-conditions: User logged in and estate selected					
Testing steps: 1. Login to the system as manager 2. Enter climate & nutrient data 3. Click “Predict Potential Yield” button.					
Test ID	Test Inputs	Expected Output	Actual Output	Result (Pass/Fail)	Comments
MID_Test_02	Provide the selected estate, target month and year, and all relevant historical climatic, nutrient, agronomic, and past yield data	Baseline target (kg), Potential target (kg), Confidence interval	Baseline target (kg), Potential target (kg), Confidence interval	Pass	Targets generated with seasonal adjustments

Table 4.3: Test Case 3

Test case ID: MID_Test_03			Test designed by: Waduthanthri W.T.H.D		
Test title: Yield Log Book Scanning and Data Extraction			Test designed day: 2025/8/28		
Test priority (High/Medium/Low): High			Test executed by: Waduthanthri W.T.H.D		
Module name: Yield Log Book Scanning and Data Extraction Module			Test executed day: 2025/12/28		
Description: Scan manually written tea yield log book images and extract recorded data into a digital format (e.g., Excel or CSV) for further analysis and integration with the predictive system.					
Pre-conditions: User logged in and has access to scanned log book images.					
Testing steps: 1. Login 2. Navigate to the “Log Book Scanning” module. 3. Upload scanned images of tea yield log books. 4. Click “Extract Data”. 5. Review extracted data preview before saving to digital format (Excel/CSV).					
Test ID	Test Inputs	Expected Output	Actual Output	Result (Pass/Fail)	Comments
MID_Test_03	Uploaded log book images (JPEG, PNG)	Extracted data displayed in structured digital format (Excel/CSV)	Extracted data successfully displayed and downloadable	Pass	User can scan handwritten records and convert them into usable digital data for analysis

5.BUDGET AND BUDGET JUSTIFICATION (IF ANY)

The estimated budget for this project primarily covers expenses related to data collection and system deployment. Traveling costs are required to collect estate-level records and environmental data from regional tea research institutes, plantations, and other relevant institutions. Additionally, monthly website hosting charges are included to maintain the online forecasting platform, ensuring continuous access for stakeholders. Since the system is built with open-source frameworks and available computing resources, no significant costs are anticipated for commercial software or specialized hardware.

Table 5.1: Budget of the entire project per month

Component	Amount (USD)	Amount (LKR)
Traveling expenses for data collection and estate visits (per month)	11.67	3,500.00
Website hosting charges (per month)	16.00	4,800.00
Total	27.67	8,300.00

6. COMMERCIALIZATION

The proposed **T-AI system** is an AI-powered platform designed to support tea estate operations. One of its main modules, the **Tea Yield Prediction and Logbook Data Extraction System**, analyzes historical data to provide accurate yield forecasts and actionable insights. This module is a scalable, data-driven decision-support tool for estate managers, smallholders, and agronomists. It also helps preserve manually recorded logbooks by digitizing them, protecting valuable historical data and making it easier to access and analyze.

The system will be offered through a tiered subscription model, from affordable packages for smallholders to full enterprise solutions for large estates. It helps reduce yield uncertainties, optimize management practices, and prevent resource misallocation.

Strategic partnerships with the Tea Smallholders Development Authority, Sri Lanka Tea Board, and regional agricultural extension services will help establish credibility and support adoption. Marketing will include workshops, field demonstrations, and digital outreach.

Over time, the platform aims to become a standard decision-support tool for sustainable tea cultivation, with potential for regional expansion and continuous improvement based on user feedback and technological advances.

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6.1 Commercial Value and Future Impact

Brand Differentiation: Use of the T-AI system, including the Predictive Analytics for Tea Crop Yield module helps estates adopt data-driven decision-making, improving yield accuracy and production planning, which strengthens their competitive position in the market.

Sustainability Impact: By forecasting yields more precisely, the module reduces fertilizer and resource waste, supports energy efficiency, and promotes eco-friendly tea production.

Farmer Empowerment: Provides smallholders and estate managers with clear yield insights, helping them make fair and transparent decisions while improving operational efficiency.



Figure 6.1 : Logo

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